

# Apéry Numbers under Exponentiation and Series Reversion

Proofs of two OEIS supercongruence conjectures  
A self-contained framing argument for A267220

ChatGPT

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## Abstract

Let  $a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$  be the classical Apéry numbers, let  $G(z) = \exp(\sum_{n \geq 1} a_n z^n / n)$ , and put  $F(w) = w / \text{Rev}(zG(z))(w)$ . We prove both conjectures recorded by Peter Bala in OEIS A267220 on 17 October 2024: for every integer  $m$ , the sequences  $u_m(n) = [z^n]G(z)^{mn}$  and  $v_m(n) = [w^n]F(w)^{mn}$  satisfy order-two prime-power supercongruences. The proof extends both conjectures to  $p = 3$ , and extends the second to  $p = 2$  as well. It also gives parameter-dependent divisibility and proves integrality of the normalized family  $B_t(n) = t^{-1}[z^n]G(z)^{tn}$ , with  $B_0(n) = a_n$ . An elementary binomial-product argument supplies the initial congruences. An explicit Frobenius-defect identity then establishes the required change-of-variable theorem, a rational-coefficient instance of the known framing principle for 2-functions. The special behavior at 2 is tracked without suppressing a denominator. Small exact examples show that a uniform cubic modulus is false, and that the first family cannot be extended unchanged to 2. A standard-library Python program provides independent finite checks using integer and rational arithmetic.

**Scope and status.** This article supplies mathematical proofs, not an inference from numerical experiments. The general framing principle is prior work, not a novelty claim. The specific OEIS applications and refinements are proved explicitly here. This is an unrefereed research exposition; no proof-assistant verification or OEIS submission is claimed.

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# 1 The problem and its mathematical setting

The Apéry numbers associated with  $\zeta(3)$  are

$$a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \quad (n \geq 0). \quad (1.1)$$

They begin 1, 5, 73, 1445, 33001, 819005, 21460825, ... and form OEIS A005259 [1]. Their role in Apéry's irrationality proof and their classical supercongruences make them a natural, substantial example rather than an artificially chosen sequence; see [5] for the arithmetic background.

Define the formal power series

$$L(z) = \sum_{n \geq 1} \frac{a_n}{n} z^n, \quad G(z) = \exp L(z), \quad W(z) = \sum_{n \geq 1} \frac{a_n}{n^2} z^n. \quad (1.2)$$

We use  $G$  for the generating function called  $A$  in OEIS A267220, to avoid confusing it with the Apéry numbers themselves. If  $h(w)$  is the unique compositional inverse of  $zG(z)$ , set

$$h(w) = \text{Rev}(zG(z))(w), \quad F(w) = \frac{w}{h(w)}. \quad (1.3)$$

Here reversion means composition, not the multiplicative reciprocal. The first coefficients are

$$\begin{aligned} G(z) &= 1 + 5z + 49z^2 + 685z^3 + 11807z^4 + 232771z^5 + \cdots, \\ F(w) &= 1 + 5w + 24w^2 + 200w^3 + 2430w^4 + 36096w^5 + \cdots. \end{aligned}$$

In the entry's formula section, Peter Bala's contribution dated 17 October 2024 states the following two conjectures [2]. For each integer  $m$ , let

$$u_m(n) = [z^n]G(z)^{mn}, \quad v_m(n) = [w^n]F(w)^{mn}. \quad (1.4)$$

For every prime  $p \geq 5$  and positive integers  $n, r$ , the proposed formulas are

$$u_m(np^r) \equiv u_m(np^{r-1}) \pmod{p^{2r}}, \quad (1.5)$$

$$v_m(np^r) \equiv v_m(np^{r-1}) \pmod{p^{2r}}. \quad (1.6)$$

Both remain labeled conjectures in the entry accessed on 20 September 2026. These are the targets of the present article. An older, different integrality comment in that entry already cites a proof; it is not being presented here as an open problem.

The exponent in (1.4) grows with the index, so these are not merely fixed powers of one integral series. Reversion adds a nonlinear operation. Furthermore, the original parameter range includes negative integers,  $m = 0$ , and parameters divisible by the modulus prime. A proof by division by  $m$  or  $m - 1$  in a finite field would miss some of those cases.

The appropriate organizing object is a *2-function*, a formal series whose coefficients obey a prime-by-prime divisibility condition. Schwarz, Vologodsky, and Walcher established preservation of this structure under framing, a change of variable involving compositional inversion [3, 4]. Our proof develops exactly the local rational-coefficient calculation needed here. In particular, it keeps the unsigned convention dictated by (1.4) and treats its behavior at 2 directly.

## 2 Main results and conventions

All series manipulations take place formally, initially over  $\mathbb{Q}$ . No analytic convergence is assumed. We write  $\text{ord}_p(q)$  for the exponent of a prime  $p$  in a nonzero rational number  $q$  and use  $\text{ord}_p(0) = +\infty$ . Thus  $q \in \mathbb{Z}_p$  means  $\text{ord}_p(q) \geq 0$ ; this is simply a convenient way of saying that the reduced denominator of  $q$  is not multiple of  $p$ . A congruence between integers has its usual meaning. For a variable  $z$  put  $\delta_z = z d/dz$ .

For an integer  $t$  and  $n \geq 1$ , define

$$B_t(n) = \begin{cases} \frac{1}{t} [z^n] G(z)^{tn}, & t \neq 0, \\ a_n, & t = 0. \end{cases} \quad (2.1)$$

The value at zero is the removable value: the coefficient of  $t$  in  $[z^n] \exp(tnL(z))$  is  $a_n$ . More generally  $B_t(n)$  is a rational polynomial in  $t$  of degree at most  $n - 1$ .

**Theorem 2.1** (Normalized supercongruences). *For every integer  $t$  and every  $n \geq 1$ ,  $B_t(n)$  is an integer. For an odd prime  $p$  and an integer  $N \geq 1$  divisible by  $p$ ,*

$$B_t(N) \equiv B_t(N/p) \pmod{p^{2\text{ord}_p(N)}}. \quad (2.2)$$

*If  $N$  is even, then*

$$\text{ord}_2(B_t(N) - B_t(N/2)) \geq \begin{cases} 2\text{ord}_2(N), & t \text{ even}, \\ 2\text{ord}_2(N) - 1, & t \text{ odd}. \end{cases} \quad (2.3)$$

**Theorem 2.2** (Both OEIS conjectures, with refinements). *For all integers  $m$  and positive integers  $n$ ,*

$$u_m(n) = mB_m(n), \quad v_m(n) = mB_{m-1}(n). \quad (2.4)$$

*Consequently, (1.5) and (1.6) hold for every odd prime, including 3. In fact, if  $m \neq 0$ ,  $p$  is odd, and  $p \mid N$ , then*

$$\text{ord}_p(u_m(N) - u_m(N/p)) \geq 2\text{ord}_p(N) + \text{ord}_p(m), \quad (2.5)$$

$$\text{ord}_p(v_m(N) - v_m(N/p)) \geq 2\text{ord}_p(N) + \text{ord}_p(m). \quad (2.6)$$

*The second OEIS congruence also holds for  $p = 2$ , for every integer  $m$ . More precisely, for nonzero  $m$  and even  $N$ ,*

$$\text{ord}_2(u_m(N) - u_m(N/2)) \geq \begin{cases} 2\text{ord}_2(N) + \text{ord}_2(m), & m \text{ even}, \\ 2\text{ord}_2(N) - 1, & m \text{ odd}, \end{cases} \quad (2.7)$$

$$\text{ord}_2(v_m(N) - v_m(N/2)) \geq \begin{cases} 2\text{ord}_2(N) + \text{ord}_2(m) - 1, & m \text{ even}, \\ 2\text{ord}_2(N), & m \text{ odd}. \end{cases} \quad (2.8)$$

*For  $m = 0$ , both sequences are zero at positive indices, so all these congruences are automatic.*

Taking  $N = np^r$  gives  $\text{ord}_p(N) = r + \text{ord}_p(n) \geq r$ , explaining why the stated results imply precisely the moduli requested in the OEIS entry, even when  $p \mid n$ . The two identities in (2.4) also show that the apparently different conjectures belong to one normalized family, with parameters differing by one.

The proof has three ingredients. First, a binomial-product identity establishes order-two congruences for the seed  $a_n$  at every prime. Second, these congruences make  $W$  a 2-function and force an integral Euler product for  $G$ . Finally, a change of variable preserves the relevant Frobenius divisibility. Lagrange inversion identifies its coefficients with  $B_t(n)$ .

### 3 An elementary congruence for the Apéry seed

It is known that Apéry numbers satisfy the stronger order-three congruences at primes at least 5; the historical results of Gessel and Coster are reviewed in [5]. We do not need that stronger theorem. The following direct argument proves all the input used below, including at 2 and 3.

**Lemma 3.1** (Binomial product). *For integers  $1 \leq k \leq N$ , put  $C(N, k) = \binom{N}{k} \binom{N+k}{k}$ . Then*

$$C(N, k) = (-1)^{k-1} \frac{N(N+k)}{k^2} \prod_{j=1}^{k-1} \left(1 - \frac{N^2}{j^2}\right). \quad (3.1)$$

*If  $p \mid N$  and  $1 \leq \ell \leq N/p$ , then*

$$C(N, p\ell) = (-1)^{(p-1)\ell} C(N/p, \ell) \prod_{\substack{1 \leq j < p\ell \\ p \nmid j}} \left(1 - \frac{N^2}{j^2}\right). \quad (3.2)$$

*Proof.* Extract  $N/k$  from  $\binom{N}{k}$  and  $(N+k)/k$  from  $\binom{N+k}{k}$ . Pairing the remaining factors gives  $(N-j)(N+j)/j^2 = -(1 - N^2/j^2)$  and proves (3.1). When  $k = p\ell$ , the factors with  $p \mid j$  form exactly the product in (3.1) for  $(N/p, \ell)$ . The prefactors coincide. The ratio of signs is  $(-1)^{p\ell-1-(\ell-1)} = (-1)^{(p-1)\ell}$ , proving (3.2).  $\square$

**Proposition 3.2** (Order two at every prime). *For every prime  $p$  and every positive  $N$  divisible by  $p$ ,*

$$a_N \equiv a_{N/p} \pmod{p^{2\operatorname{ord}_p(N)}}. \quad (3.3)$$

*Proof.* Write  $r = \operatorname{ord}_p(N)$ . If  $p \nmid k$ , the identity  $\binom{N}{k} = (N/k) \binom{N-1}{k-1}$  gives  $\operatorname{ord}_p(\binom{N}{k}) \geq r$ . Hence  $C(N, k)^2$  is divisible by  $p^{2r}$ .

If  $k = p\ell$ , each denominator  $j$  in the remaining product of (3.2) is a  $p$ -adic unit. Thus each factor is 1 modulo  $p^{2r}$ . On squaring, the sign disappears, and we obtain

$$C(N, p\ell)^2 \equiv C(N/p, \ell)^2 \pmod{p^{2r}}.$$

This holds also for  $p = 2$ : no assertion that the unsquared sign is positive has been made. The term  $k = 0$  is 1 on both sides. Summing over all  $k$  proves (3.3).  $\square$

*Remark 3.3.* One must split the product before reducing it modulo a prime power. Factors with  $p \mid j$  need not be congruent to 1 and cannot be discarded. Their exact contribution is the lower-index binomial product.

### 4 Dilogarithms, Euler products, and local defects

We now isolate the general arithmetic structure. The subsequent argument works for any integer sequence satisfying (3.3), not just the Apéry numbers.

**Definition 4.1.** An integer sequence  $(a_n)_{n \geq 1}$  will be called *order-two admissible* if (3.3) holds for every prime and every positive multiple  $N$  of that prime. Its associated series  $W$  in (1.2) is then a rational 2-function.

**Proposition 4.2** (Three equivalent descriptions). *For an integer sequence  $(a_n)_{n \geq 1}$ , the following are equivalent:*

- (i) *The sequence is order-two admissible.*
- (ii) *For every  $d \geq 1$ ,*

$$c_d = \frac{1}{d^2} \sum_{e|d} \mu(e) a_{d/e} \in \mathbb{Z}. \quad (4.1)$$

- (iii) *For every prime  $p$ ,*

$$R_p(z) := \frac{1}{p^2} W(z^p) - W(z) \in z\mathbb{Z}_p[[z]]. \quad (4.2)$$

When these conditions hold,

$$W(z) = \sum_{d \geq 1} c_d \operatorname{Li}_2(z^d), \quad (4.3)$$

$$G(z) = \prod_{d \geq 1} (1 - z^d)^{-dc_d} \in 1 + z\mathbb{Z}[[z]]. \quad (4.4)$$

*Proof.* Suppose (i), fix  $d = p^r h$  with  $p \nmid h$ , and group terms in the numerator of (4.1) according to whether the squarefree divisor  $e$  contains  $p$ . The numerator becomes

$$\sum_{e|h} \mu(e) (a_{p^r h/e} - a_{p^{r-1} h/e}).$$

Every summand is divisible by  $p^{2r}$ . Doing this for every prime dividing  $d$  proves divisibility by  $d^2$ , hence (ii).

Möbius inversion gives

$$a_n = \sum_{d|n} d^2 c_d. \quad (4.5)$$

Conversely, if (ii) holds and  $N = p^r h$  with  $p \nmid h$ , subtracting the expressions (4.5) for  $N$  and  $N/p$  leaves only divisors  $d$  with  $\operatorname{ord}_p(d) = r$ . Their squares are multiples of  $p^{2r}$ , proving (i).

For (iii), the coefficient of  $z^N$  in  $R_p$  is

$$[z^N] R_p(z) = \begin{cases} -a_N/N^2, & p \nmid N, \\ (a_{N/p} - a_N)/N^2, & p \mid N. \end{cases} \quad (4.6)$$

Since the  $a_N$  are integers, the equivalence of (i) and (iii) follows. Finally, expand  $\operatorname{Li}_2(z) = \sum_{j \geq 1} z^j/j^2$  and apply (4.5) to obtain (4.3). Because  $\delta_z \operatorname{Li}_2(z^d) = -d \log(1 - z^d)$ , exponentiation gives (4.4). All products and sums are locally finite as formal series. Integer, including negative, powers of an integral unit have integer coefficients, which proves the last assertion.  $\square$

For the Apéry seed,  $c_1, \dots, c_6$  are 5, 17, 160, 2058, 32760, 596092. For example, the first factors of  $G$  are

$$(1 - z)^{-5} (1 - z^2)^{-34} (1 - z^3)^{-480} (1 - z^4)^{-8232} \dots$$

Their integrality is a consequence of the congruences, not an assumption based on the initial coefficient list.

The logarithmic derivative of (4.2) is also integral:

$$D_p(z) := \delta_z R_p(z) = \frac{1}{p} L(z^p) - L(z) \in z\mathbb{Z}_p[[z]]. \quad (4.7)$$

Although  $W$  and  $L$  can have denominators, all higher derivatives needed below are integral:

$$\delta_z^k W(z) = \sum_{n \geq 1} n^{k-2} a_n z^n \in z\mathbb{Z}[[z]] \quad (k \geq 2). \quad (4.8)$$

This distinction between the first derivative and the higher derivatives is important in the cancellation argument.

## 5 The normalized family as a change of variable

For an integer  $t$ , introduce

$$f_t(z) = zG(z)^{-t}, \quad \psi_t(w) = \text{Rev}(f_t)(w). \quad (5.1)$$

Since  $G$  is an integral unit,  $f_t(z) = z + O(z^2)$  has integer coefficients. Its inverse  $\psi_t(w) = w + O(w^2)$  has integer coefficients as well: the coefficient of degree  $n$  in the equation  $f_t(\psi_t(w)) = w$  determines the new inverse coefficient with coefficient 1, using only integer polynomials in earlier coefficients.

Set

$$W_t(w) = W(\psi_t(w)) - \frac{t}{2} L(\psi_t(w))^2. \quad (5.2)$$

The quadratic correction is essential. Merely composing  $W$  with  $\psi_t$  does not yield the cancellation used later.

**Lemma 5.1** (Differential and coefficient identities). *For every integer  $t$ ,*

$$\delta_w W_t(w) = L(\psi_t(w)), \quad (5.3)$$

$$W_t(w) = \sum_{n \geq 1} \frac{B_t(n)}{n^2} w^n. \quad (5.4)$$

*Proof.* In the coordinate  $w = zG(z)^{-t}$ ,

$$\frac{dw}{w} = (1 - t\delta_z L(z)) \frac{dz}{z}, \quad d\left(W(z) - \frac{t}{2} L(z)^2\right) = L(z)(1 - t\delta_z L(z)) \frac{dz}{z}.$$

The factor  $1 - t\delta_z L(z)$  has constant term 1, so is a formal unit. Division gives (5.3).

The Lagrange coefficient formula, proved in [section A](#), says that if  $\psi(w) = w\Phi(\psi(w))$ , then for  $n \geq 1$ ,

$$[w^n]H(\psi(w)) = \frac{1}{n} [z^{n-1}]H'(z)\Phi(z)^n. \quad (5.5)$$

Here  $\Phi = G^t$  and  $H = L$ . When  $t \neq 0$ ,

$$\begin{aligned} [w^n]L(\psi_t(w)) &= \frac{1}{n} [z^{n-1}]L'(z)G(z)^{tn} \\ &= \frac{1}{tn^2} [z^{n-1}](G(z)^{tn})' \\ &= \frac{1}{tn} [z^n]G(z)^{tn} = \frac{B_t(n)}{n}. \end{aligned}$$

For  $t = 0$ ,  $\psi_0(w) = w$  and the same identity follows from  $B_0(n) = a_n$ . Both sides of (5.4) have zero constant coefficient, so logarithmic integration proves it.  $\square$

At this point  $B_t(n)$  is only known to be rational when  $t \neq 0$ . The next section proves the needed local integrality before using it to assert divisibility. This avoids treating division by  $t$  as though it were harmless at a prime dividing  $t$ .

## 6 The Frobenius-defect calculation

This section gives the central proof. It uses only the conclusions of [proposition 4.2](#), the integral coordinate inverse, and elementary bounds on factorial denominators.

### 6.1 An integral logarithmic error

Fix a prime  $p$  and an integer  $t$ . Write

$$x = \psi_t(w), \quad x_p = \psi_t(w^p), \quad J(w) = G(\psi_t(w)) \in 1 + w\mathbb{Z}[[w]]. \quad (6.1)$$

Define

$$S(w) = \frac{1}{p} \log J(w^p) - \log J(w) = \frac{1}{p} L(x_p) - L(x). \quad (6.2)$$

**Lemma 6.1.** *For every prime  $p$ , including 2,  $S(w) \in w\mathbb{Z}_p[[w]]$ . Moreover,*

$$x_p = x^p \exp(ptS(w)). \quad (6.3)$$

*Proof.* Reduction modulo  $p$  gives  $J(w^p) \equiv J(w)^p$  because the coefficients are integers. Thus

$$\frac{J(w^p)}{J(w)^p} = 1 + pH(w), \quad H(w) \in w\mathbb{Z}_p[[w]].$$

The formal logarithm yields

$$S(w) = \frac{1}{p} \log(1 + pH(w)) = \sum_{j \geq 1} \frac{(-1)^{j+1} p^{j-1}}{j} H(w)^j.$$

The scalar coefficients are  $p$ -integral, since  $j - 1 - \text{ord}_p(j) \geq 0$  for every  $j \geq 1$ . This includes  $p = 2$ . For example, if  $e = \text{ord}_p(j)$  then  $j \geq 2^e \geq e + 1$ . The sum is well defined coefficientwise because  $H$  has zero constant term.

The inverse-coordinate equation says  $x = wJ(w)^t$ . Applying it at  $w^p$  and dividing by  $x^p = w^p J(w)^{pt}$  gives

$$\frac{x_p}{x^p} = \left( \frac{J(w^p)}{J(w)^p} \right)^t = \exp(ptS(w)),$$

which is (6.3). All quotients here are quotients of formal units after the common leading monomial has been canceled.  $\square$

### 6.2 The exact cancellation identity

The framed Frobenius defect is

$$\tilde{R}_{p,t}(w) = \frac{1}{p^2} W_t(w^p) - W_t(w). \quad (6.4)$$

The goal is to prove its coefficients integral, allowing at most one factor of 2 in their denominators when both  $p = 2$  and  $t$  is odd.

**Proposition 6.2** (Explicit defect formula). *With the notation above,*

$$\boxed{\begin{aligned} \tilde{R}_{p,t}(w) &= R_p(x) + tS(w)D_p(x) - \frac{t}{2}S(w)^2 \\ &\quad + \sum_{k \geq 2} \frac{p^{k-2}t^k}{k!} S(w)^k (\delta_z^k W)(x^p). \end{aligned}} \quad (6.5)$$

*This is an equality of formal series over  $\mathbb{Q}_p$ .*

*Proof.* For a series  $V$  with zero constant coefficient, the logarithmic Taylor identity is

$$V(ye^q) = \sum_{k \geq 0} \frac{q^k}{k!} (\delta_y^k V)(y). \quad (6.6)$$

It follows by checking the monomials  $y^n$ : both sides are  $y^n e^{nq}$ . Afterward one may substitute series for  $y$  and  $q$  having zero constant term. Every coefficient then involves only finitely many terms.

Apply (6.6) to  $W$ , using  $y = x^p$  and  $q = ptS$ :

$$\begin{aligned} \frac{1}{p^2} W(x_p) &= \frac{1}{p^2} W(x^p) + \frac{tS}{p} L(x^p) \\ &\quad + \sum_{k \geq 2} \frac{p^{k-2}t^k}{k!} S^k (\delta_z^k W)(x^p). \end{aligned}$$

Also, by (6.2),  $L(x_p)/p = L(x) + S$ . The quadratic part of (5.2) therefore contributes

$$-\frac{t}{2} \left( \left( \frac{L(x_p)}{p} \right)^2 - L(x)^2 \right) = -tSL(x) - \frac{t}{2}S^2.$$

Combining the linear Taylor term with the first term of this expression gives

$$tS \left( \frac{L(x_p)}{p} - L(x) \right) = tSD_p(x).$$

The terms involving  $W$  with no Taylor derivatives give  $R_p(x)$ . These substitutions prove (6.5).  $\square$

The terms  $L(x)$  and  $L(x^p)/p$  need not be integral separately. Their cancellation is what prevents the denominator  $n$  in the logarithm from costing a power of the modulus. After this cancellation, only integral series and explicit factorial denominators remain.

### 6.3 Factorial denominators and the prime two

**Lemma 6.3.** *For  $k \geq 2$  and an odd prime  $p$ ,*

$$\text{ord}_p(k!) \leq k - 2. \quad (6.7)$$

*For  $p = 2$ ,*

$$\text{ord}_2(k!) \leq k - 1. \quad (6.8)$$

*Consequently  $p^{k-2}/k!$  is  $p$ -integral for odd  $p$ , and  $2^{k-2}/k! \in \frac{1}{2}\mathbb{Z}_2$ .*

*Proof.* Counting multiples of each power of  $p$  among  $1, \dots, k$  gives  $\text{ord}_p(k!) = \sum_{j \geq 1} \lfloor k/p^j \rfloor$ . Equivalently, if  $s_p(k) \geq 1$  is the sum of the base- $p$  digits of  $k$ , this is  $(k - s_p(k))/(p - 1)$ . For odd  $p$  and  $k \geq 3$ , it is at most  $(k - 1)/2 \leq k - 2$ ;  $k = 2$  is immediate. At  $p = 2$  it is at most  $k - 1$ .  $\square$

**Theorem 6.4** (Unsigned framing, with the 2-adic loss). *Let  $(a_n)_{n \geq 1}$  be any order-two admissible integer sequence, and define  $L, G, W, \psi_t, W_t$  as above. For every integer  $t$ ,*

$$\tilde{R}_{p,t}(w) \in \begin{cases} w\mathbb{Z}_p[[w]], & p \text{ odd}, \\ w\mathbb{Z}_2[[w]], & p = 2 \text{ and } t \text{ even}, \\ \frac{1}{2}w\mathbb{Z}_2[[w]], & p = 2 \text{ and } t \text{ odd}. \end{cases} \quad (6.9)$$

*Its normalized coefficients  $B_t(n)$  are integers for all integers  $t$ , and they satisfy (2.2) and (2.3).*

*Proof.* By [proposition 4.2](#) and [lemma 6.1](#), the series  $R_p(x)$ ,  $D_p(x)$ , and  $S$  are  $p$ -integral. By (4.8), so is  $(\delta_z^k W)(x^p)$  for every  $k \geq 2$ . Composition preserves integrality because  $x \in w\mathbb{Z}[[w]]$ .

For odd  $p$ , the number  $1/2$  is  $p$ -integral, and [lemma 6.3](#) makes every scalar  $p^{k-2}t^k/k!$  integral. Thus (6.5) lies in  $w\mathbb{Z}_p[[w]]$ .

For  $p = 2$ , the same argument allows at worst a factor  $1/2$ , both from  $tS^2/2$  and from the factorial coefficients. If  $t$  is even, the former term is integral and so are the latter coefficients: for nonzero even  $t$  and  $k \geq 2$ ,

$$\text{ord}_2 \left( \frac{2^{k-2}t^k}{k!} \right) \geq k - 2 + k - (k - 1) = k - 1 \geq 1.$$

For  $t = 0$  all those terms vanish. This proves (6.9).

We now deduce the coefficient assertions without assuming in advance that  $B_t(n)$  is integral. From (5.4),

$$[w^N]\tilde{R}_{p,t}(w) = \begin{cases} -B_t(N)/N^2, & p \nmid N, \\ (B_t(N/p) - B_t(N))/N^2, & p \mid N. \end{cases} \quad (6.10)$$

When the defect is integral, the first case shows that  $B_t(N)$  is  $p$ -integral for  $p \nmid N$ . The second case then proves, by induction on  $\text{ord}_p(N)$ , that every  $B_t(N)$  is  $p$ -integral; simultaneously it proves that  $B_t(N) - B_t(N/p)$  has valuation at least  $2\text{ord}_p(N)$ . This covers every odd prime and also  $p = 2$  when  $t$  is even.

For odd  $t$ , the definition (2.1) has numerator in  $\mathbb{Z}$ , since  $G^{tn}$  is an integer power of an integral unit, and denominator  $t$  is odd. Hence  $B_t(n)$  is 2-integral in that case as well. The half-integral defect in (6.10) gives exactly the lower bound  $2\text{ord}_2(N) - 1$  for the difference. The case  $t = 0$  is also immediate from  $B_0(n) = a_n$ . Thus every rational  $B_t(n)$  is integral at every prime, and therefore belongs to  $\mathbb{Z}$ . All the asserted bounds follow.  $\square$

[Proposition 3.2](#) supplies the hypotheses for Apéry numbers, so [theorem 6.4](#) proves [theorem 2.1](#). The proof makes no coprimality assumption between  $t$  and  $p$ . Its only use of the ordinary quotient by  $t$  for integrality is at  $p = 2$  when  $t$  is odd, where that quotient is legitimate.

## 7 Finishing both conjectures

The first identity in (2.4) is immediate for  $m \neq 0$ , and for  $m = 0$  both sides are zero at positive indices. The reversion identity requires a separate coefficient calculation.

**Proposition 7.1** (A parameter shift accounts for reversion). *For every integer  $m$  and every  $n \geq 1$ ,*

$$[w^n]F(w)^{mn} = mB_{m-1}(n).$$

*In particular,  $[w^n]F(w)^n = a_n$ .*

*Proof.* The inverse  $h(w)$  in (1.3) is  $\psi_{-1}(w)$ . The equation  $w = h(w)G(h(w))$  gives

$$F(w) = G(h(w)) \in 1 + w\mathbb{Z}[[w]].$$

Apply (5.5) to  $h = wG(h)^{-1}$  and  $H(z) = G(z)^{mn}$ . Since integer powers, including negative powers, are well defined formally,

$$\begin{aligned} [w^n]F(w)^{mn} &= \frac{1}{n}[z^{n-1}](G(z)^{mn})'G(z)^{-n} \\ &= m[z^{n-1}]L'(z)G(z)^{(m-1)n}. \end{aligned}$$

For  $m \neq 1$ , differentiating  $G^{(m-1)n}$  turns the last expression into

$$\frac{m}{m-1}[z^n]G(z)^{(m-1)n} = mB_{m-1}(n).$$

When  $m = 1$ , the previous line before division is  $[z^{n-1}]L'(z) = a_n = B_0(n)$ . The case  $m = 0$  gives zero directly. These are equalities over  $\mathbb{Q}$ , so no modular division by  $m - 1$  is used.  $\square$

*Proof of theorem 2.2.* The exact identities (2.4) have now been established. For odd  $p$ , multiply (2.2) by  $m$  to obtain (2.5) and (2.6) for  $m \neq 0$ . They imply the original congruences, and the same proof applies at  $p = 3$ .

For  $u_m$  at 2, the framing parameter is  $t = m$ . If  $m$  is even, (2.3) gives full precision  $2 \operatorname{ord}_2(N)$ , and multiplication by  $m$  supplies  $\operatorname{ord}_2(m)$  more powers. If  $m$  is odd, it gives precision  $2 \operatorname{ord}_2(N) - 1$  and multiplication loses or gains no power of 2. This is (2.7).

For  $v_m$  the framing parameter is  $m - 1$ . If  $m$  is odd, that parameter is even, so (2.3) gives full precision. If  $m$  is nonzero even, the normalized family can lose one power of 2, but the prefactor  $m$  supplies at least one. More precisely the resulting bound is  $2 \operatorname{ord}_2(N) + \operatorname{ord}_2(m) - 1$ . Thus (2.8) holds, and its lower bound is at least  $2 \operatorname{ord}_2(N)$  in both cases. This proves the second OEIS congruence at 2 for every  $m$ , including the trivial case  $m = 0$ .  $\square$

The value  $v_1(n) = a_n$  recovers the familiar sequence rather than creating an unrelated one. Another useful exact relation is

$$v_2(n) = 2B_1(n) = 2u_1(n). \tag{7.1}$$

More generally, if  $m \neq 1$ ,  $v_m(n) = m u_{m-1}(n)/(m - 1)$  as a rational identity. The normalized form is preferable for divisibility arguments because the integrality of  $B_{m-1}(n)$  has already been proved even at primes dividing  $m - 1$ .

## 8 Examples and limits of the result

### 8.1 Integer-valued parameter polynomials

The first few normalized polynomials are

$$\begin{aligned} B_t(1) &= 5, \\ B_t(2) &= 50t + 73, \\ B_t(3) &= \frac{1125t^2 + 3285t + 2890}{2}, \\ B_t(4) &= \frac{20000t^3 + 87600t^2 + 147574t + 99003}{3}. \end{aligned}$$

Their coefficients need not be integers. The theorem asserts that their values at every integer  $t$  are integers. This is a genuine integer-valued-polynomial conclusion, not a statement that all coefficient denominators disappear algebraically.

For completeness, expanding the exponential gives the finite expression

$$B_t(n) = \sum_{\substack{k_1, \dots, k_n \geq 0 \\ k_1 + 2k_2 + \dots + nk_n = n}} n^K t^{K-1} \prod_{j=1}^n \frac{(a_j/j)^{k_j}}{k_j!}, \quad K = k_1 + \dots + k_n. \quad (8.1)$$

Since  $n \geq 1$ , every summand has  $K \geq 1$ . At  $t = 0$  only  $K = 1$  survives, and that term is  $a_n$ . Formula (8.1) is useful for symbolic experimentation, but its denominators do not by themselves explain either integrality or the supercongruences.

| $t$ | $B_t(1)$ | $B_t(2)$ | $B_t(3)$ | $B_t(4)$ | $B_t(5)$ |
|-----|----------|----------|----------|----------|----------|
| -2  | 5        | -27      | 410      | -1915    | 53130    |
| -1  | 5        | 23       | 365      | 6343     | 123255   |
| 0   | 5        | 73       | 1445     | 33001    | 819005   |
| 1   | 5        | 123      | 3650     | 118059   | 4015380  |
| 2   | 5        | 173      | 6980     | 301517   | 13540505 |
| 3   | 5        | 223      | 11435    | 623375   | 35175630 |

Table 1: Exact values, including negative framing parameters.

### 8.2 The obstruction at two is real

For  $m = 1$ ,

$$u_1(1) = 5, \quad u_1(2) = [z^2]G(z)^2 = 2 \cdot 49 + 5^2 = 123.$$

Therefore

$$u_1(2) - u_1(1) = 118 \equiv 2 \pmod{4}. \quad (8.2)$$

So an assertion that the first conjecture holds for all primes and all integers  $m$  would be false. The lower bound  $2 \operatorname{ord}_2(2) - 1 = 1$  in (2.7) is attained. In contrast,  $v_2(2) - v_2(1) = 2 \cdot 118 = 236$  is divisible by 4, illustrating the compensating factor in the second family. This does not claim that every individual odd parameter always fails the stronger 2-adic modulus; it disproves the uniform assertion.

### 8.3 A uniform cubic modulus is false

The initial Apéry numbers have order-three congruences for  $p \geq 5$ , but a general framing need not preserve that order. Exact coefficient extraction gives

$$u_1(7) = 5\,082\,313\,276,$$

and hence

$$\begin{aligned} u_1(7) - u_1(1) &= 5\,082\,313\,271 \\ &= 7^2 \cdot 103\,720\,679, \quad 103\,720\,679 \equiv 6 \pmod{7}. \end{aligned}$$

Thus its 7-adic valuation is exactly 2, not 3, and its residue modulo 343 is 294. By (7.1),

$$v_2(7) - v_2(1) = 10\,164\,626\,542 \equiv 245 \pmod{343},$$

again with valuation exactly 2. Therefore neither conjecture admits a uniform replacement of  $p^{2r}$  by  $p^{3r}$  for all parameters, even in the original range  $p \geq 5$ . These are counterexamples to stronger variants, not to either formula actually listed in OEIS.

## 9 A broader family covered by the same argument

The transform theorem is not specific to one recurrence. It applies to any integer seed satisfying (3.3). There is also a direct extension of the elementary binomial argument.

**Proposition 9.1.** *For every positive integer  $h$ , the binomial-moment sequence*

$$a_n^{(h)} = \sum_{k=0}^n \left( \binom{n}{k} \binom{n+k}{k} \right)^{2h}$$

*is order-two admissible. Consequently its associated normalized family satisfies theorem 2.1, and its exponential and reverted families satisfy theorem 2.2 with this seed in place of  $a_n$ .*

*Proof.* If  $p \nmid k$ , the argument of proposition 3.2 gives  $\text{ord}_p(C(N, k)^{2h}) \geq 2h \text{ord}_p(N) \geq 2 \text{ord}_p(N)$ . If  $k = p\ell$ , raise (3.2) to the even power  $2h$ . The sign disappears and each remaining factor is 1 modulo  $p^{2 \text{ord}_p(N)}$ . Summing proves the required seed congruence, after which theorem 6.4 and proposition 7.1 apply without change.  $\square$

The case  $h = 1$  is exactly the classical Apéry sequence used to resolve the two selected conjectures. This extension is included to show what the proof depends on: an arithmetic property of a binomial sum and an integral change of variable, not a numerical fit or an unproved classification of recurrences.

## 10 Exact computation and reproducibility

The accompanying program `verify.py` uses only the Python standard library. It uses arbitrary-precision integers and `Fraction`; no floating-point equality or probabilistic primality decision enters the checks. The following describes the saved run with limit 150.

## 10.1 Two ways to generate the seed

For efficiency, the program generates the seed by the classical recurrence [5]

$$(n+1)^3 a_{n+1} = (2n+1)(17n^2 + 17n + 5)a_n - n^3 a_{n-1}, \quad a_0 = 1, a_1 = 5. \quad (10.1)$$

It independently evaluates the defining binomial sum at every index  $0 \leq n \leq 150$  and checks equality. Every division in the recurrence is checked for zero remainder. The mathematical proof in this article uses the binomial definition, not the recurrence, so no separate proof of (10.1) is an unmentioned dependency of the main theorem.

## 10.2 Exact coefficients and independent reversion

For a fixed integer exponent  $q$ , write  $G(z)^q = \sum_{j \geq 0} h_j z^j$ . Logarithmic differentiation gives

$$h_0 = 1, \quad j h_j = q \sum_{k=1}^j a_k h_{j-k} \quad (j \geq 1). \quad (10.2)$$

To compute  $B_t(n)$ , the program sets  $q = tn$ , evaluates the recurrence through degree  $n$ , and divides by  $t$  for  $t \neq 0$ . All remainders are checked; for  $t = 0$  the seed value is used.

Independently, the program constructs the compositional inverse of  $zG(z)$  by solving its coefficients successively, checks the composition, forms  $F = w/h(w)$ , and computes the powers  $F^{mn}$  by rational polynomial convolution. Those tests do not use  $v_m(n) = mB_{m-1}(n)$  to generate the left-hand side. They therefore provide a separate check of the parameter shift and the reversion convention.

Finally, the full two sides of (6.5) are constructed as truncated rational series through degree 14. This checks the quadratic sign, the factor  $p^{k-2}$ , and the placement of  $x^p$  as distinct from  $x_p$ , rather than only comparing a selection of final congruences.

| Check performed in the saved run                        | Count |
|---------------------------------------------------------|-------|
| Binomial-sum versus recurrence values, indices 0–150    | 151   |
| Seed congruences, all primes and multiples $N \leq 150$ | 269   |
| Odd-prime normalized congruences on the specified grid  | 798   |
| Normalized 2-adic bounds, all even $N \leq 150$         | 1575  |
| Parameter-sensitive odd-prime bounds for $u_m$          | 840   |
| Parameter-sensitive 2-adic bounds for $u_m$             | 1575  |
| Bounds for $v_m$ , including the prime 2                | 2415  |
| Independent reversion-and-power identities              | 126   |
| Full Frobenius-defect identities through degree 14      | 18    |

Table 2: All listed exact checks passed. These finite counts are validation, not a substitute for the general proof.

The normalized odd-prime grid uses  $t = -10, -9, \dots, 10$ ,  $p \in \{3, 5, 7, 11, 13\}$ ,  $1 \leq r \leq 4$ , and  $1 \leq n \leq 4$ , retaining  $N = np^r \leq 150$ . The 2-adic normalized tests use all even  $N \leq 150$  for the same  $t$  range. The parameter-sensitive  $u_m$  and odd-prime  $v_m$  tests use  $m = -10, \dots, 10$  and multiples  $N \leq 50$  of those odd primes; the  $u_m$  and  $v_m$  tests additionally check their full parameter-sensitive bounds at every even  $N \leq 150$ . The independent reversion tests use  $m = -4, \dots, 4$  and  $1 \leq n \leq 14$ . The 18 formal defect tests use  $t \in \{-3, -1, 0, 1, 2, 4\}$  and  $p \in \{2, 3, 5\}$ . The sharpness examples in section 8 are also checked exactly.

### 10.3 Rebuilding the artifacts

From the archive directory, the verification command is

```
python3 verify.py --out data --limit 150
```

Run Python normally, not with the `-O` option: the verification uses assertions. The output includes a JSON summary and CSV tables. The PDF can be rebuilt without a separate bibliography processor:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The README records the artifact inventory. Numerical checks cannot prove an infinite family of congruences. Their role here is to make the examples reproducible and to expose implementation, normalization, or sign errors in independently computed quantities. The proofs establish the claims for every parameter and index, beyond the finite test range.

## 11 Conclusion and attribution

Both October 2024 supercongruence conjectures in A267220 are true. The same argument strengthens the first from  $p \geq 5$  to every odd prime, and the second to every prime. It proves integrality of the normalized family, gains powers from the integer parameter, and identifies the precise general one-power loss at 2 for odd unsigned framings. Counterexamples show why two natural stronger uniform assertions must not be inferred.

The structural explanation is that the Apéry binomial sum produces a 2-function, and the two OEIS constructions read off coefficients after closely related changes of variable. The known framing theory of [3, 4] is therefore the appropriate context. The article has not asserted that this general principle, the classical Apéry congruences, or the old integrality remarks in OEIS are new. Nor does an entry's conjectural label establish that no proof exists elsewhere. What is supplied here is an explicit self-contained derivation of the two listed claims, with the prime-2 refinement and all integer-parameter edge cases accounted for.

## A A formal proof of the Lagrange formula

We include the coefficient formula to make the power-series part of the argument self-contained. Let  $\Phi(z) \in 1 + z\mathbb{Q}[[z]]$ , put  $f(z) = z/\Phi(z)$ , and let  $\psi$  be its inverse. For a Laurent series,  $\text{Res}_z$  means the coefficient of  $z^{-1}$  in a formal one-form. Formal residue is invariant under an invertible change of variable, so

$$\begin{aligned} [w^n]H(\psi(w)) &= \text{Res}_w H(\psi(w))w^{-n-1}dw \\ &= \text{Res}_z H(z)f(z)^{-n-1}f'(z)dz. \end{aligned}$$

For clarity, residue invariance can be checked on each monomial  $w^j$ : if  $j \neq -1$ , the substituted form is the derivative of  $f(z)^{j+1}/(j+1)$  and has zero residue; if  $j = -1$ ,  $f'/f$  has residue 1 because  $f = z$  times a unit.

Using  $d(f^{-n}) = -nf^{-n-1}f' dz$  and the fact that a formal derivative has zero residue, integration by parts gives

$$\begin{aligned} \operatorname{Res}_z H(z)f(z)^{-n-1}f'(z) dz &= \frac{1}{n} \operatorname{Res}_z H'(z)f(z)^{-n} dz \\ &= \frac{1}{n} [z^{n-1}] H'(z)\Phi(z)^n. \end{aligned}$$

This is (5.5), for all positive  $n$ . The constant term of  $H$  is irrelevant to both sides.

## B Proof dependencies and boundary cases

For an independent audit, the logical dependencies can be summarized without treating computation as a lemma. The binomial identity proves [proposition 3.2](#). Möbius inversion proves [proposition 4.2](#), including integrality of  $G$ . Integral compositional inversion and the Lagrange formula identify the coefficients of  $W_t$ . The elementary logarithmic lemma, exact Taylor cancellation, and factorial bounds prove [theorem 6.4](#). The parameter-shift identity then proves both OEIS statements.

The points most vulnerable to an incorrect shortcut are the following. The inverse is *compositional*, and  $F = G \circ \psi_{-1}$ , not  $G \circ \psi_1$ . The quadratic correction has sign  $-t/2$  for the coordinate  $w = zG^{-t}$ . The Frobenius Taylor expansion uses  $x^p$ , whereas  $x_p$  denotes  $\psi_t(w^p)$ ; these are generally different. The normalized coefficient  $B_t(n)$  is not assumed integral before its local proof. Division by  $m-1$  is avoided at  $m=1$  and never used modulo  $p$ . Negative parameters are integer powers of units and need no positivity assumption. The two-adic half-integrality is retained rather than silently replaced by integrality. Finally, the modulus depends on the full valuation of  $N$ , not on a representation of  $N$  with a tacitly prime-to- $p$  leading factor.

At index zero the original definitions give  $u_m(0) = v_m(0) = 1$ , even for  $m=0$ . All normalized identities in this article are stated for positive indices. The conjectures themselves have positive  $n$  and  $r$ , so this convention causes no exceptional case in their application.

## References

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